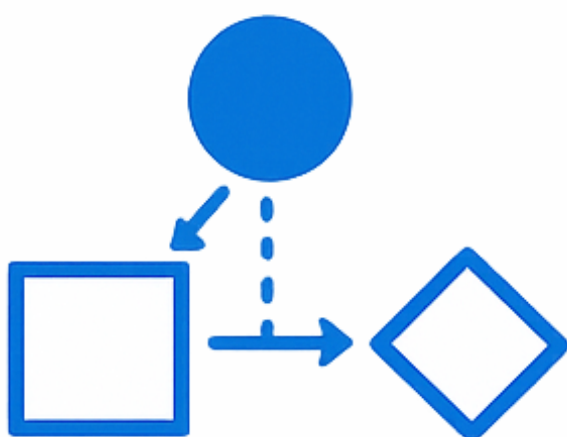




## Assignment 2

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## Qualitative Methods and Data Quality

Part I: Say-Do Gap Study and Part II: Radiology Data Quality Analysis

Date: March 2, 2026

Course: NDAB20010U Health Data and Interoperability

|                               |                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------|
| Institution:                  | University of Copenhagen, Faculty of Health and Medical Sciences                      |
| Department:                   | Department of Public Health                                                           |
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| Course:                       | NDAB20010U Health Data and Interoperability                                           |
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| Date:                         | March 2, 2026                                                                         |

## AI Declaration

I declare that AI has only been used for grammatical correction.

AI has **not** been used to generate content, reasoning, or structure.

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# Abbreviation Key

Table 1: Internal abbreviations and identifiers used in this report

| Abbreviation | Meaning in this report                                                             |
|--------------|------------------------------------------------------------------------------------|
| Axx          | Activity identifier in the interview-based model in Part I (for example, A01–A09). |
| Exx          | Event identifier in the observed Part I event log (for example, E03, E10).         |
| Dxx          | Delta identifier in the Part I comparison sheet (for example, D01–D06).            |
| Mx           | Method identifier in the Part II data quality toolkit (for example, M2, M10).      |
| RQ1          | Research Question 1: workflow efficiency and bottlenecks.                          |
| RQ2          | Research Question 2: timeliness of critical finding communication.                 |
| RQ3          | Research Question 3: AI-based triage system development.                           |
| DQ           | Data quality.                                                                      |
| TAT          | Turnaround time between selected workflow milestones.                              |

# 1 Part I: A Real-World Study of the Say-Do Gap in Modelling

This part analyses one weekday morning routine across *work-as-imagined*, *work-as-done*, and *work-as-recorded*, with explicit links between claims and evidence.

## 1.1 Deliverable 1: Description of the Routine Based on the Interview

**Task Requirement (Deliverable 1)** *Describe the routine from the interview and identify activities, their order, and participants.*

**Scope and evidence** This Part I study is a single-case self-study with one participant (the author) and one routine (weekday morning preparation before leaving home). The interview was conducted on February 18, 2026 as a semi-structured self-interview.

Following Chapter 27, this step is treated as *manual knowledge acquisition*: it captures *knowledge-as-stated* (what the participant reports) and produces an initial *work-as-imagined* model (how the routine is expected to unfold) (Guide to Health Informatics 2026; Zajac 2026). Internal identifiers used in the report are summarised in Table 1.

The routine boundary is anchored in transcript evidence (Appendix A, Table 3, lines L21–L22):

- Start: the alarm sounds.
- End: the apartment door closes when leaving.

The baseline sequence is grounded in lines L02–L12: wake-up, hygiene, dressing/packing, breakfast, final checks, and departure. Variation points (snooze, breakfast substitution, key search, occasional roommate coordination) are grounded in lines L14–L18.

**Participants, order, and modelling assumptions** Table 4 presents the interview-based activity model, and Figure 1 shows the corresponding interview-based DCR graph. In this context, interview-based DCR means a process model built from interview evidence rather than direct observation.

Assumptions used in Deliverable 1 are:

- The model represents a weekday baseline, not weekend behaviour.
- The participant is the only consistent human actor in the routine.
- Expected times are recall-based estimates and therefore approximate.

Together, these assumptions keep scope explicit and make later comparison with observed evidence in Deliverables 2 and 3 auditable.

## 1.2 Deliverable 2: Event Log Representing the Observed Routine

**Task Requirement (Deliverable 2)** *Create an event log of the observed routine as a CSV file with timestamps, event labels, and relevant context notes.*

The observation was conducted on February 20, 2026 (06:45–07:30), two days after the interview. This delay was intentional to reduce recall carry-over between elicitation and observation.

In methodological terms, the observation captures *work-as-done* (what was actually performed in context). The resulting CSV captures *work-as-recorded* (what is represented in structured trace data) (Guide to Health Informatics 2026; Zajac 2026). To keep data collection transparent, the observation followed a short guide (Appendix A, Table 2).

The submitted event log includes the fields `case_id`, `event_id`, `timestamp`, `event_label`, `actor`, `tool`, `context_note`, and `related_activity_id`. The full machine-readable file is submitted separately, while Appendix A, Table 5 shows the key excerpt used in the report.

Compared with the interview baseline, the observation reveals three anchor differences that are used in Deliverable 3:

- E03: extra coordination event,
- E09: order shift relative to the interview sequence,
- E10: explicit workaround behaviour.

These observations provide the empirical basis for the delta analysis between stated and observed practice.

## 1.3 Deliverable 3: Delta Sheet Summarising Differences Between Stated and Observed Practice

**Task Requirement (Deliverable 3)** *Compare Deliverables 1 and 2 and produce a delta sheet with missing events, extra events, different order, timing distortions, workarounds/coordination steps, and relabeling/ambiguity.*

The comparison maps interview activities (Axx identifiers) to observed events (Exx identifiers). A delta is registered when at least one of the following differs: event presence, order, timing, workaround behaviour, or activity interpretation. This follows the Chapter 27 logic of comparing observed traces with a prior process description to expose model limits and recurring exceptions (Guide to Health Informatics 2026).

Appendix A, Table 6 contains the complete delta sheet. The six required categories are all represented:

- Missing event: no distinct seated breakfast-and-schedule-review step.

- Extra event: roommate coordination appears only in observed data.
- Different order: bag packing occurs after breakfast preparation starts.
- Timing distortion: hygiene begins later than the interview baseline.
- Workaround/coordination step: key/lock issue triggers ad hoc messaging.
- Relabeling/ambiguity: *breakfast* changes from sit-down meal to banana plus coffee-to-go.

These deltas operationalise the say-do gap, meaning the difference between described routine and performed routine. They also prepare the reflection in Deliverable 4 by showing where model assumptions break under real execution conditions.

## 1.4 Deliverable 4: Reflections on the Modelling Process

**Task Requirement (Deliverable 4)** *Reflect on six modelling questions on missed data, activity definition, event-log limits, automation potential, observed variations, and the role of qualitative discovery.*

1. **Question 1.** *What kinds of data were missed or misrepresented in the interview-based model and why?*

### Answer

**Claim.** The interview-based model captured intent, but underrepresented situated coordination and recovery behaviour.

**Evidence.** Observation added micro-coordination (E03), workaround behaviour (E10), and omission of the planned seated breakfast-and-schedule step.

**Implication.** Interview-only modelling overstates routine linearity and should be treated as a starting model, not a full process description.

2. **Question 2.** *What counts as an activity in each model, and how did this differ?*

### Answer

**Claim.** The two models use different granularity.

**Evidence.** Deliverable 1 uses intention-level activities (Axx), while Deliverable 2 logs timestamped observable events (Exx).

**Implication.** One interview activity can map to multiple observed events, so apparent stability at intention level can hide operational variation.

3. **Question 3.** *How did the event log constrain what could be modelled?*

**Answer**

**Claim.** The event log supports sequence and timing analysis well, but limits interpretation of motives and local judgement.

**Evidence.** The log records *what* happened and *when*, while causal context is only partially recoverable from notes.

**Implication.** Process modelling from logs alone can identify deviations, but cannot fully explain why deviations occurred.

4. **Question 4.** *Do you think that the event log could be captured automatically by a system? Why or why not?*

**Answer**

**Claim.** Automatic capture is feasible for digital traces but incomplete for social and tacit work.

**Evidence.** Timestamped app events can be captured reliably, but ad hoc coordination and workaround decisions are often off-system.

**Implication.** Automated logging should be combined with targeted qualitative follow-up when model validity depends on context.

5. **Question 5.** *How did the observation help uncover variations or exceptions that were not mentioned in the interview?*

**Answer**

**Claim.** Observation exposed exceptions that were absent from the stated routine.

**Evidence.** The observed trace showed order changes (E07 before E09), a missing distinct A06 step, and workaround handling (E10).

**Implication.** Conformance checking should be paired with exception analysis, because recurring local deviations can be practically rational.

6. **Question 6.** *Why is it relevant to consider qualitative methods of discovery (e.g., observations) when building process models?*

**Answer**

**Claim.** Qualitative methods improve model validity by grounding formal traces in real practice.

**Evidence.** The combined use of interview, observation, and event log produced a more coherent account than any single source alone.



**Implication.** Robust process models require triangulation across stated, observed, and recorded evidence (Guide to Health Informatics 2026; Zajac 2026).

## 2 Part II: Radiology Case Study — Data Quality, Conformance, and the Limits of Logs

This part synthesises notebook evidence (Deliverable 5) and the accompanying Part II reflection section (Deliverable 6).

### 2.1 Deliverable 5: Notebook Submission

**Task Requirement (Deliverable 5)** *Submit the completed notebook `assignment.ipynb` with the full Part II workflow and method execution.*

The completed notebook is submitted as a separate artefact (`assignment.ipynb`) in the required folder.

It includes all Part II steps (Parts 0–7): initial exploration, selection and execution of data quality (DQ) methods, custom DQ dimension design, per-facet synthesis, and invisible-work analysis.

Method identifiers (Mx) and research-question identifiers (RQ1–RQ3) follow the abbreviation key in Table 1.

### 2.2 Deliverable 6: Part II Reflection Questions

1. **Initial exploration. What kind of data is logged in the file? Assuming data quality is good, how can the data be used to answer the RQs?**

#### **Answer**

**Claim.** The file is a case-based radiology event log with timestamps, activity labels, actor/role information, and optional context fields.

**Evidence.** The profile in Appendix B, Table 7, shows event-level traces across 17 cases, and Table 12 lists the logged workflow activities.

**Implication.** If data quality were high, the same structure could answer all research questions: RQ1 through turnaround and queue analysis, RQ2 through critical communication conformance checks, and RQ3 through feature extraction for triage-model development.

2. **Method selection justification. Include your completed method selection table from Part 3. Additionally, answer why you did not choose the other methods, what trade-offs were considered, and how RQs influenced selection.**

#### **Answer**

**Claim.** The selected set (M2, M6, M8, M10, M11) was chosen to maximise direct relevance to RQ1–RQ3 while limiting analytical overlap.

**Evidence.** The completed selection table is included in Appendix B, Table 8. In short:

- M2 establishes baseline completeness before interpretation.
- M6 tests representativity across urgency and temporal context.
- M8 directly addresses patient-safety communication timeliness (RQ2).
- M10 connects performance analysis (RQ1) to timestamp credibility.
- M11 adds workload-distribution context relevant to RQ1 and RQ3.

Methods not selected were not irrelevant; they were deprioritised. M1 and M3 were narrower for the chosen question set, M4 and M5 partly overlapped selected outputs, M7 overlapped timeliness analysis already covered by M10, and M9 was clinically useful but less central for cross-RQ synthesis.

**Implication.** The key trade-off was breadth versus depth: five methods provided enough coverage for all RQs without fragmenting interpretation.

3. **Results interpretation. Provide detailed interpretation of results from each chosen method (key results, interpretation, expectation check, and primary facets).**

**Answer**

**Claim.** The selected methods produced interpretable and complementary evidence, with the strongest risk signal related to temporal credibility.

**Evidence.** Key outputs are summarised in Appendix B, Table 9, with supporting visualisations in Figures 2–5.

**M2 (Completeness).**

**Key result.** Several context fields have high missingness.

**Interpretation.** Sequence-level analysis remains possible, but richer contextual interpretation is constrained.

**Expectation check.** Expected direction confirmed.

**Primary facets.** Data, Source, System.

**M6 (Representativity).**

**Key result.** Strong urgency and shift imbalance.

**Interpretation.** The observed data are not evenly representative across operational contexts.

**Expectation check.** Expected, but skew magnitude was larger than anticipated.

**Primary facets.** Data, Task, Human.

#### **M8 (Critical communication conformance).**

**Key result.** Most critical cases pass, with one failed communication case.

**Interpretation.** Even a single failed critical communication event is clinically meaningful.

**Expectation check.** Near-complete compliance was expected; the failed case remains a material safety signal.

**Primary facets.** Task, System, Human.

#### **M10 (TAT with batch-logging detection).**

**Key result.** Raw and cleaned TAT estimates differ materially; 35.29% of cases were batch-logging flagged.

**Interpretation.** Operational performance conclusions are sensitive to whether timestamps reflect real-time entry.

**Expectation check.** Partly expected; effect size was the main surprise.

**Primary facets.** Data, System, Task.

#### **M11 (Workload distribution).**

**Key result.** Clear concentration across roles and actors.

**Interpretation.** Human resource patterns materially shape what the log appears to show about efficiency and model readiness.

**Expectation check.** Concentration was expected and confirmed.

**Primary facets.** Human, Task, System.

**Implication.** The methods worked as intended, but interpretation quality depends on combining metrics with provenance and practice context.

#### **4. Your proposed DQ dimension. Define the dimension, include code, explain motivation with concrete examples, RQ impact, and facet mapping.**

##### **Answer**

**Claim.** The proposed custom dimension is *operational authenticity*: whether records represent real clinical workflow rather than test or administrative traces.

**Evidence.** Notebook Part 5 identified a high-risk outlier (TEST-001) with multiple non-operational markers. Summary results are in Appendix B.2, Table 10. The implemented method is documented in Appendix B.3, Listing 1. Facet mapping is in Table 11.

**Implication.** This issue affects all RQs: it can bias performance estimates (RQ1), distort safety conformance denominators (RQ2), and contaminate model development data (RQ3).

5. **Per-Facet Data Quality Synthesis.** For each facet, assess findings, main issues, and consequences for RQs. Then discuss the most challenging facet and alternative assessments.

**Answer**

**Claim.** Dataset quality is uneven across the five facets (Data, Source, System, Task, Human), with the strongest challenges in Human and Source interpretation.

**Evidence.** This synthesis draws on Appendix B, Table 9, the custom-dimension tables in Appendix B.2, and Figures 2–5.

**Data facet.** Missingness and timestamp-credibility sensitivity increase measurement risk for RQ1 and RQ3.

**Source facet.** Mixed provenance and contamination signals reduce confidence in generalisation, especially for RQ3.

**System facet.** Logging design shapes what is visible; communication and timestamp capture limits constrain RQ1 and RQ2 conclusions.

**Task facet.** Urgency/shift skew and rule-dependent interpretation suggest partial alignment with real priorities.

**Human facet.** Workload concentration and documentation behaviour influence the observed trace and can bias interpretation.

**Most challenging facet.** Human (and closely Source), because event logs reveal actions but rarely intent, negotiation, or informal coordination.

**Alternative assessment options.**

- Add entry-mode and provenance metadata to logs.
- Run targeted short interviews for high-impact anomalies.
- Link across systems (for example, RIS/PACS/communication traces).
- Use periodic documentation-practice audits.

**Implication.** Per-facet judgement is strongest when quantitative methods are paired with focused qualitative validation.

6. **Invisible work and system design implications.** Use the Part 7 analysis and workflow diagram to address missing activities, reasons, system-design consequences, Part I parallel, and one concrete logging-system change with trade-offs.

**Answer**

**Claim.** The event log omits clinically relevant coordination work, and this omission has direct design consequences for all three RQs.

**Evidence.** The missing-activity table is provided in Appendix B.5, Table 13. Interpretation is anchored to the workflow diagram in Appendix B.4, Figure 6. Missing activities are plausibly unlogged for three reasons:

- Technical: phone/verbal interactions occur outside core schemas.
- Social: coordination is often informal and situational.
- Economic/workload: exhaustive documentation increases burden.

If a process model is built only from event-log data, RQ1 bottlenecks can be misattributed, RQ2 communication pathways can look safer than practice, and RQ3 models can learn incomplete process representations. This mirrors Part I: interview versus observation gaps parallel workflow-diagram versus event-log gaps.

**Implication.** Data-driven healthcare design is necessary but insufficient on its own. One concrete improvement is a lightweight `communication/coordination` event with minimal required fields (timestamp, role, linked case, reason code). The trade-off is documentation burden and surveillance sensitivity; mitigation is targeted capture at high-impact points rather than full universal logging (Guide to Health Informatics 2026; Guide to Health Informatics 2026).

## A Supporting Tables for Part I

This appendix contains supplementary Part I material referenced from Deliverables 1–4.

Table 2: Observation guide used during Deliverable 2

| Focus                              | Observation prompt                                                             | Evidence target                       |
|------------------------------------|--------------------------------------------------------------------------------|---------------------------------------|
| Sequence and boundary              | Which event marks start/end, and what is the observed order of actions?        | Event IDs, timestamps, sequence notes |
| Tools and artefacts                | Which tools/resources are used at each step?                                   | Tool field, context notes             |
| Interactions and coordination      | Does any interaction with other people occur?                                  | Actor field, coordination notes       |
| Deviations from interview baseline | Which actions are skipped, added, reordered, or relabeled relative to A01-A09? | Related activity ID + delta notes     |
| Adaptation and recovery            | Which workaround or exception handling occurs under time/resource pressure?    | Context notes for exception events    |

*Source: Zajac (2026) and Guide to Health Informatics (2026a): interview + observation + process-tracing principles adapted to this routine study.*

*Note: Field notes were split into descriptive and reflexive notes; only descriptive notes were transcribed into the submitted CSV event log.*

## A.1 Core Deliverable Artefacts (Part I)

Table 3: Line-numbered self-interview transcript (conducted February 18, 2026)

| Line | Transcript                                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------------|
| L01  | Interview question: Please describe your normal weekday morning routine from when you wake up until you leave home. |
| L02  | Participant: My alarm goes off at 06:45, and I usually get up right away after stopping it.                         |
| L03  | Participant: I first drink a glass of water, then I go to the bathroom.                                             |
| L04  | Participant: I normally take a quick shower, around ten to twelve minutes.                                          |
| L05  | Participant: After that I get dressed and prepare my bag for the day.                                               |
| L06  | Participant: I usually pack my laptop, charger, student ID card, wallet, and keys.                                  |
| L07  | Participant: Then I make coffee and toast.                                                                          |
| L08  | Participant: I normally sit down, eat breakfast, and check my calendar and transport timing on my phone.            |
| L09  | Participant: After breakfast I brush my teeth and do a final mirror check.                                          |
| L10  | Participant: I check the weather so I can choose the right jacket.                                                  |
| L11  | Participant: Before leaving, I do a final essentials check: phone, wallet, keys, and card.                          |
| L12  | Participant: I aim to leave at around 07:30 to catch transport to campus.                                           |
| L13  | Interview question: Which steps vary the most from day to day?                                                      |
| L14  | Participant: Sometimes I snooze once if I am tired.                                                                 |
| L15  | Participant: Breakfast can change; if I am in a rush, I might just eat fruit instead of toast.                      |
| L16  | Participant: Occasionally I spend a minute looking for keys if I did not put them in the same place.                |
| L17  | Interview question: Are other people typically involved in this routine?                                            |
| L18  | Participant: Mostly no, I do it alone, but once in a while I text my roommate if I need a shared item.              |
| L19  | Interview question: Which tools or systems are central in the routine?                                              |
| L20  | Participant: My phone alarm, calendar, weather app, the coffee machine, toaster, and my backpack.                   |
| L21  | Interview question: How do you define start and end of this routine?                                                |
| L22  | Participant: It starts when the alarm sounds and ends when I close the apartment door behind me.                    |



Table 4: Interview-based activity model (work-as-imagined / knowledge-as-stated)

| ID  | Activity label                    | Actor       | Order | Expected time | Main tools/resources          |
|-----|-----------------------------------|-------------|-------|---------------|-------------------------------|
| A01 | Stop alarm and orient to time     | Participant | 1     | 06:45-06:47   | Smartphone alarm              |
| A02 | Drink water                       | Participant | 2     | 06:47-06:50   | Glass, tap                    |
| A03 | Toilet and shower                 | Participant | 3     | 06:50-07:02   | Bathroom, shower              |
| A04 | Dress and pack bag                | Participant | 4     | 07:02-07:12   | Wardrobe, backpack            |
| A05 | Prepare breakfast and coffee      | Participant | 5     | 07:12-07:20   | Coffee machine, toaster       |
| A06 | Eat breakfast and review schedule | Participant | 6     | 07:20-07:25   | Phone calendar/transport apps |
| A07 | Brush teeth and final grooming    | Participant | 7     | 07:25-07:28   | Toothbrush, mirror            |
| A08 | Check weather and essentials      | Participant | 8     | 07:28-07:30   | Weather app, keys, wallet, ID |
| A09 | Leave apartment for commute       | Participant | 9     | 07:30         | Front door, bag               |

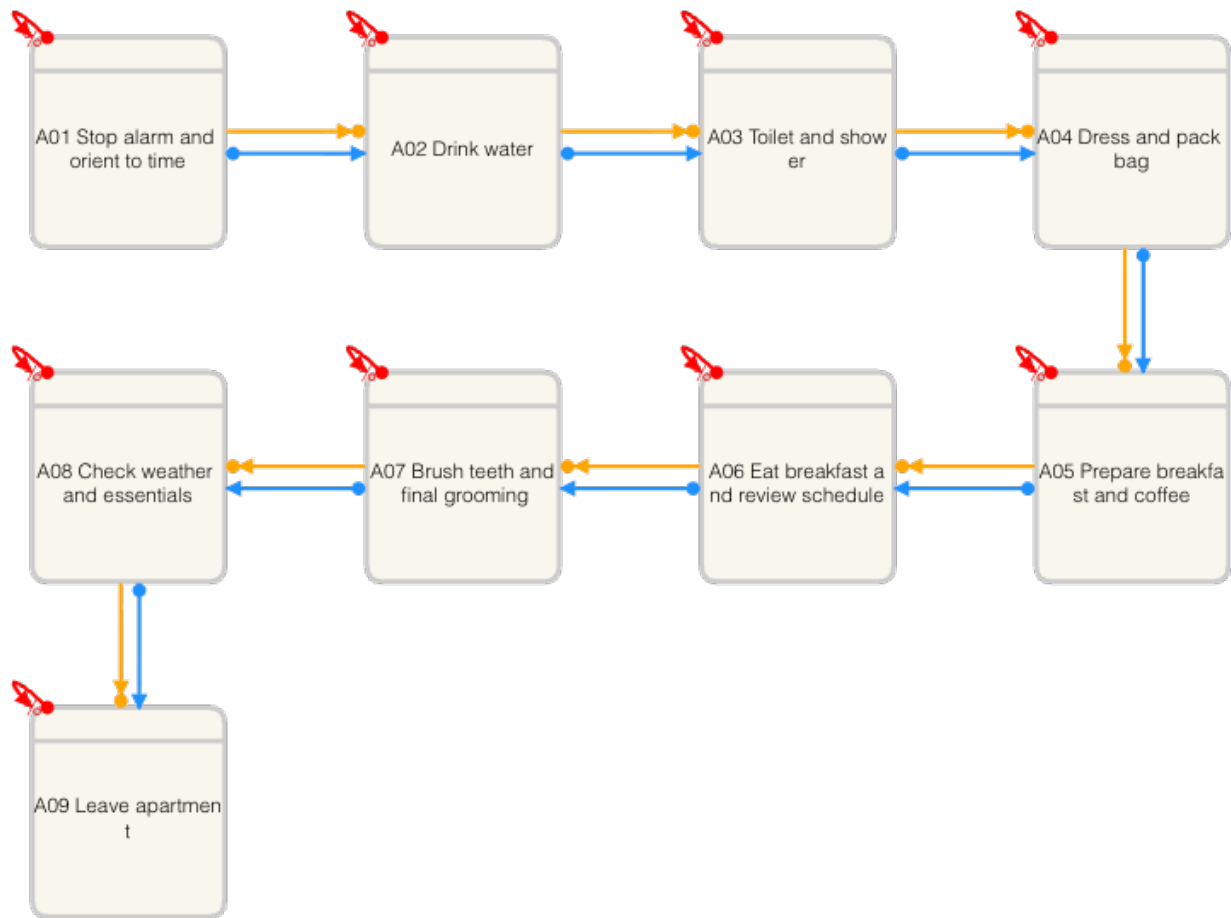


Figure 1: Interview-based DCR graph for the morning routine (A01-A09)

Source: Own DCR Graph export (February 2026), built from transcript evidence in Table 3 (lines L02-L12, L14-L18, L21-L22).

Note: Used as baseline work-as-imagined model for comparison with observed events in Deliverables 2 and 3.

Table 5: Observed event-log excerpt (February 20, 2026)

| Event | Timestamp        | Event label                | Tool                    | Context note                                            | Ref. |
|-------|------------------|----------------------------|-------------------------|---------------------------------------------------------|------|
| E01   | 2026-02-20 06:45 | Alarm rings and snooze     | Smartphone alarm        | One snooze before wake-up.                              | A01  |
| E03   | 2026-02-20 06:52 | Reply to roommate message  | Messaging app           | Extra micro-coordination event.                         | NA   |
| E05   | 2026-02-20 06:57 | Toilet and shower          | Bathroom                | Hygiene block started later than interview expectation. | A03  |
| E07   | 2026-02-20 07:12 | Start coffee and toast     | Coffee machine, toaster | Breakfast preparation begins.                           | A05  |
| E08   | 2026-02-20 07:16 | Switch breakfast to banana | Kitchen                 | Breakfast adapted due to no bread.                      | A05  |
| E09   | 2026-02-20 07:18 | Pack bag                   | Backpack                | Packing occurs after breakfast prep starts.             | A04  |
| E10   | 2026-02-20 07:21 | Find keys and borrow lock  | Keys, messaging app     | Workaround to solve missing item.                       | A08  |
| E14   | 2026-02-20 07:30 | Leave apartment            | Front door              | End of observed routine boundary.                       | A09  |

Table 6: Delta sheet between interview model and observed routine

| Delta ID | Category                       | Interview ref.           | Observed ref.   | Difference description                                                | Implication                                               |
|----------|--------------------------------|--------------------------|-----------------|-----------------------------------------------------------------------|-----------------------------------------------------------|
| D01      | Missing events                 | A06                      | No direct event | No distinct seated breakfast-and-schedule-review event occurred.      | Interview model overstates routine regularity.            |
| D02      | Extra events                   | None in baseline         | E03 (06:52)     | Short roommate-message coordination appears in observed routine only. | Reveals practical coordination load.                      |
| D03      | Different order                | A04 before A05           | E07 then E09    | Bag packing happened after breakfast preparation started.             | Ordering constraints are weaker than stated.              |
| D04      | Timing distortions             | A03 expected 06:50-07:02 | E05 start 06:57 | Hygiene block shifted later due to wake-up delays.                    | Time windows in interview are optimistic.                 |
| D05      | Workarounds/coordination steps | A08 final check          | E10 (07:21)     | Missing lock triggered ad hoc borrowing via message.                  | Recovery actions are underrepresented in interview model. |
| D06      | Relabeling/ambiguity           | A05/A06                  | E08, E13        | “Breakfast” changed from sit-down meal to banana plus coffee-to-go.   | Label definitions need operational clarity.               |

## B Supporting Tables and Code for Part II

This section includes only the relevant outputs from the completed Deliverable 5 notebook (`assignment.ipynb`): selected method evidence, key visualisations, custom DQ-dimension results, and invisible-work comparison material.

Table 7: Part II log profile from notebook Part 1 exploration

| Metric                             | Result                                                                                                                      |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Total events                       | 150                                                                                                                         |
| Unique cases                       | 17                                                                                                                          |
| Time range covered                 | 2026-01-12 07:45 to 2026-01-13 08:46                                                                                        |
| Unique logged activities           | 12                                                                                                                          |
| Columns with strongest missingness | study_uid (134), series_count (134), instance_count (134),<br>icd11_code (131), report_id (113), modality (101), notes (73) |

Table 8: Selected methods run in notebook Part 4

| Method | DQ dimension              | Primary facets       | RQ link  | Selection rationale                                     |
|--------|---------------------------|----------------------|----------|---------------------------------------------------------|
| M2     | Completeness              | Data, Source, System | RQ1–RQ3  | Baseline quality gate before downstream interpretation. |
| M6     | Representativity          | Data, Task, Human    | RQ1, RQ3 | Checks urgency and temporal balance of case coverage.   |
| M8     | Timeliness conformance    | Task, System, Human  | RQ2      | Tests safety-critical communication behavior directly.  |
| M10    | Timeliness + credibility  | Data, System, Task   | RQ1      | Quantifies TAT and sensitivity to batch logging.        |
| M11    | Workload representativity | Human, Task, System  | RQ1, RQ3 | Reveals concentration across actors/roles.              |

Table 9: Key quantitative outputs from selected methods

| Method | Key output                                                                                        | Relevance for Deliverable 6 reflection                                              |
|--------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| M2     | Context-field missingness is high (e.g., study_uid 134/150; modality 101/150).                    | Shows what can and cannot be trusted for workflow and AI-oriented interpretations.  |
| M6     | Urgency skew (15 stat/urgent, 2 routine) and shift imbalance (Day 11, Night 6, Evening 0).        | Demonstrates representativity limits for RQ1 and especially RQ3 generalisation.     |
| M8     | Critical communication: 16 pass, 1 fail (CXR-004 missing communication event).                    | Direct patient-safety signal tied to RQ2 and conformance assumptions.               |
| M10    | Mean total TAT = 147.93 min raw vs 198.50 min clean; 6/17 cases flagged as batch-logged (35.29%). | Shows strong sensitivity of performance conclusions (RQ1) to timestamp credibility. |
| M11    | Events per role: radiographer 64, system 31, senior 31, clinician 17, resident 7.                 | Quantifies workload concentration and role imbalance relevant to RQ1 and RQ3.       |

## B.1 Notebook Visualisations (Relevant Outputs)

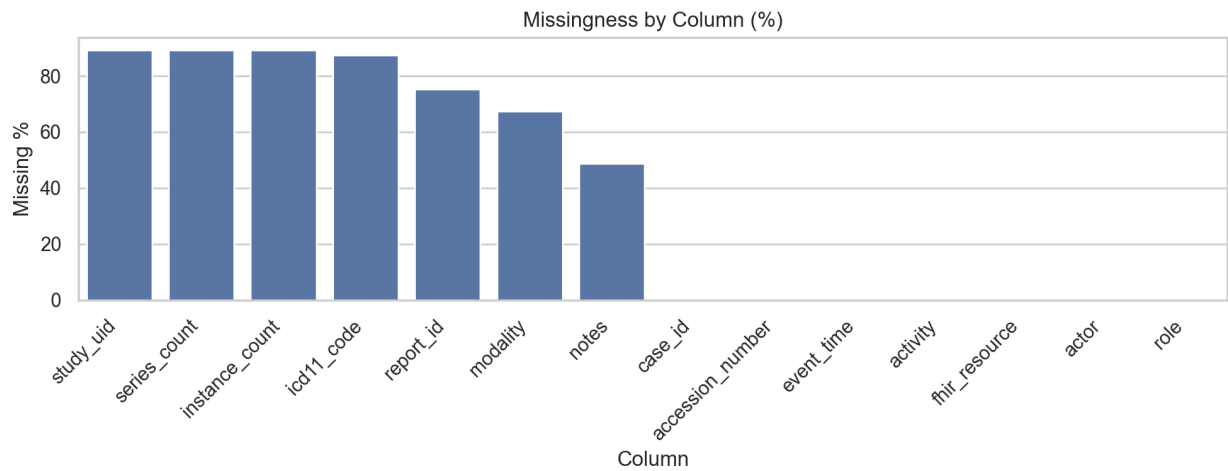


Figure 2: Method 2 output: missingness profile by column

Source: Generated from Deliverable 5 notebook output using *radiology\_event\_log.csv*.

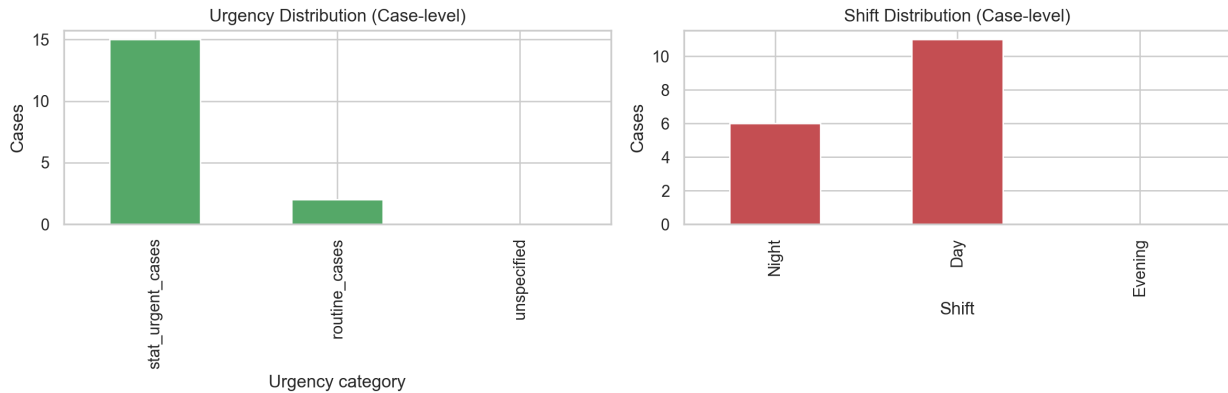


Figure 3: Method 6 output: urgency and shift representativity

Source: Generated from Deliverable 5 notebook output using *radiology\_event\_log.csv*.

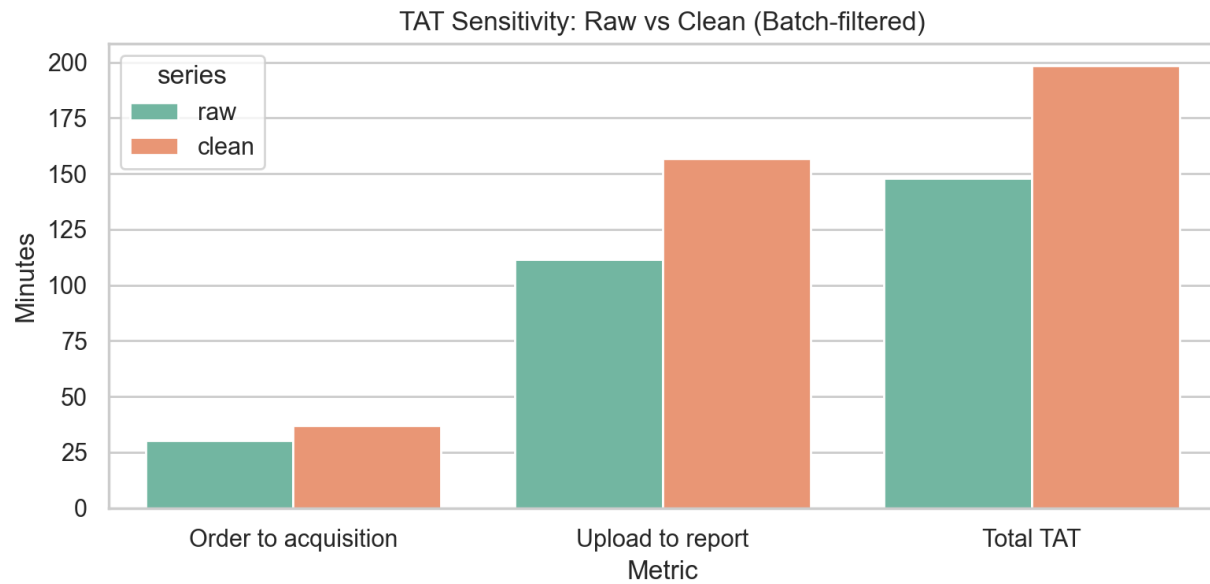


Figure 4: Method 10 output: turnaround-time sensitivity (raw vs batch-filtered)  
Source: Generated from Deliverable 5 notebook output using *radiology\_event\_log.csv*.

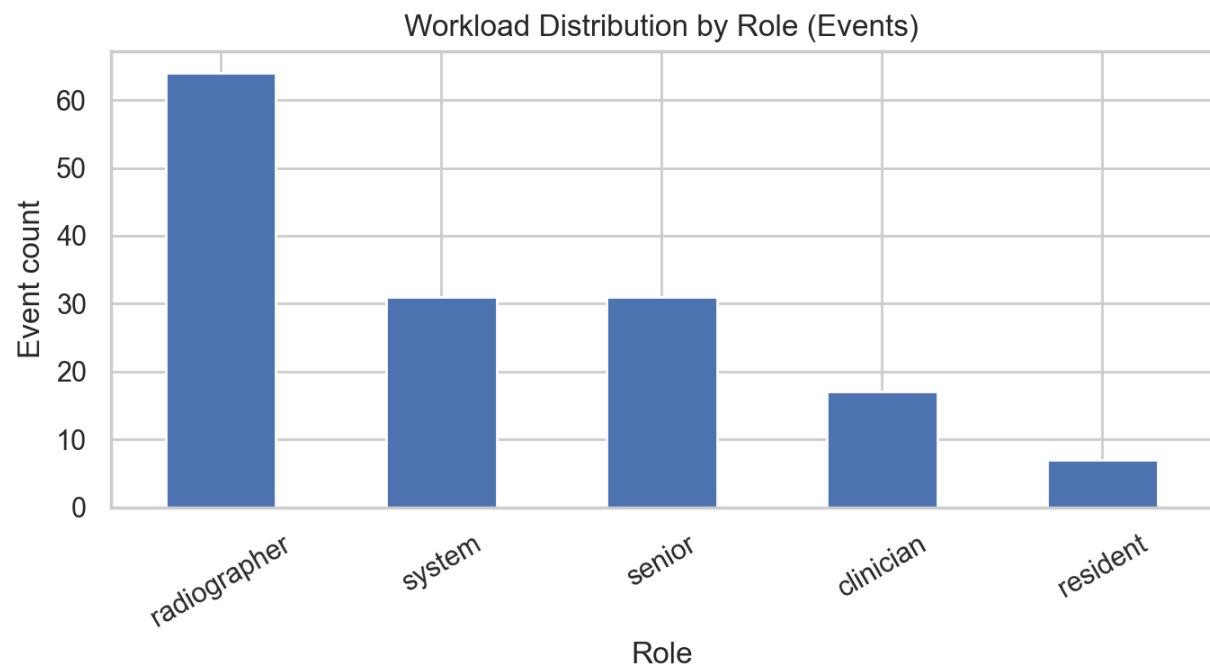


Figure 5: Method 11 output: workload distribution by role  
Source: Generated from Deliverable 5 notebook output using *radiology\_event\_log.csv*.

## B.2 Custom DQ Dimension (Notebook Part 5)

Table 10: Custom DQ dimension result summary: operational authenticity

| Metric                           | Result                                                                                                                                |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Total evaluated cases            | 17                                                                                                                                    |
| Flagged cases (medium/high risk) | 1                                                                                                                                     |
| Flagged case rate                | 5.88%                                                                                                                                 |
| High-risk cases                  | 1                                                                                                                                     |
| Flagged case details             | TEST-001: unexpected case ID format; training/test marker in notes; administrative actor involvement; extremely short trace (1 event) |

Table 11: Facet mapping for custom dimension: operational authenticity

| Facet  | Involvement | Justification                                                                               |
|--------|-------------|---------------------------------------------------------------------------------------------|
| Data   | ++          | Case identifiers and note text contain direct contamination signals (test/training traces). |
| Source | ++          | Source governance determines whether non-production traces enter the analysis dataset.      |
| System | +           | Logging architecture does not fully separate test and production provenance.                |
| Task   | +           | KPI and triage conclusions become biased when non-clinical traces are included.             |
| Human  | +           | Human setup/maintenance actions can introduce non-operational events.                       |

## B.3 Custom DQ Assessment Method Code

Listing 1: Custom DQ assessment method from notebook Part 5 (operational authenticity)

```
1 def my_custom_dq_method(df: pd.DataFrame) -> Dict[str, any]:
2     """
3     Assess operational authenticity (production-relevance validity).
4
5     Goal:
6     Identify case traces that are likely test/training/non-production records
7     or otherwise not representative of real clinical workflow data.
8     """
9     rows = []
10
11     for case_id, case_df in df.groupby('case_id'):
12         reasons = []
13
14         if not re.match(r'^CXR-\d{3}$', str(case_id)):
```



```

15         reasons.append('Unexpected case_id format')
16
17     if case_df['notes'].astype(str).str.contains(
18         r'training|test|do not process', case=False, na=False
19     ).any():
20         reasons.append('Training/test markers in notes')
21
22     if case_df['actor'].eq('Admin User').any():
23         reasons.append('Administrative/test actor involvement')
24
25     if len(case_df) < 3:
26         reasons.append('Extremely short trace (low operational realism)')
27
28     score = len(reasons)
29     if score >= 3:
30         risk_level = 'high'
31     elif score >= 1:
32         risk_level = 'medium'
33     else:
34         risk_level = 'low'
35
36     rows.append({
37         'case_id': case_id,
38         'risk_level': risk_level,
39         'risk_score': score,
40         'reasons': '; '.join(reasons) if reasons else 'None',
41         'event_count': len(case_df),
42     })
43
44     case_flags = pd.DataFrame(rows).sort_values(
45         ['risk_score', 'case_id'], ascending=[False, True]
46     )
47     flagged = case_flags[case_flags['risk_level'].isin(['high', 'medium'])].copy()
48
49     summary_metrics = {
50         'total_cases': int(len(case_flags)),
51         'flagged_cases': int(len(flagged)),
52         'flagged_case_rate_pct': round(
53             100 * len(flagged) / len(case_flags), 2
54         ) if len(case_flags) > 0 else 0.0,
55         'high_risk_cases': int((case_flags['risk_level'] == 'high').sum()),
56     }
57
58     return {
59         'case_flags': case_flags,

```

```

60     'flagged_cases': flagged,
61     'summary_metrics': summary_metrics,
62 }

```

## B.4 Workflow Diagram from Assignment Brief

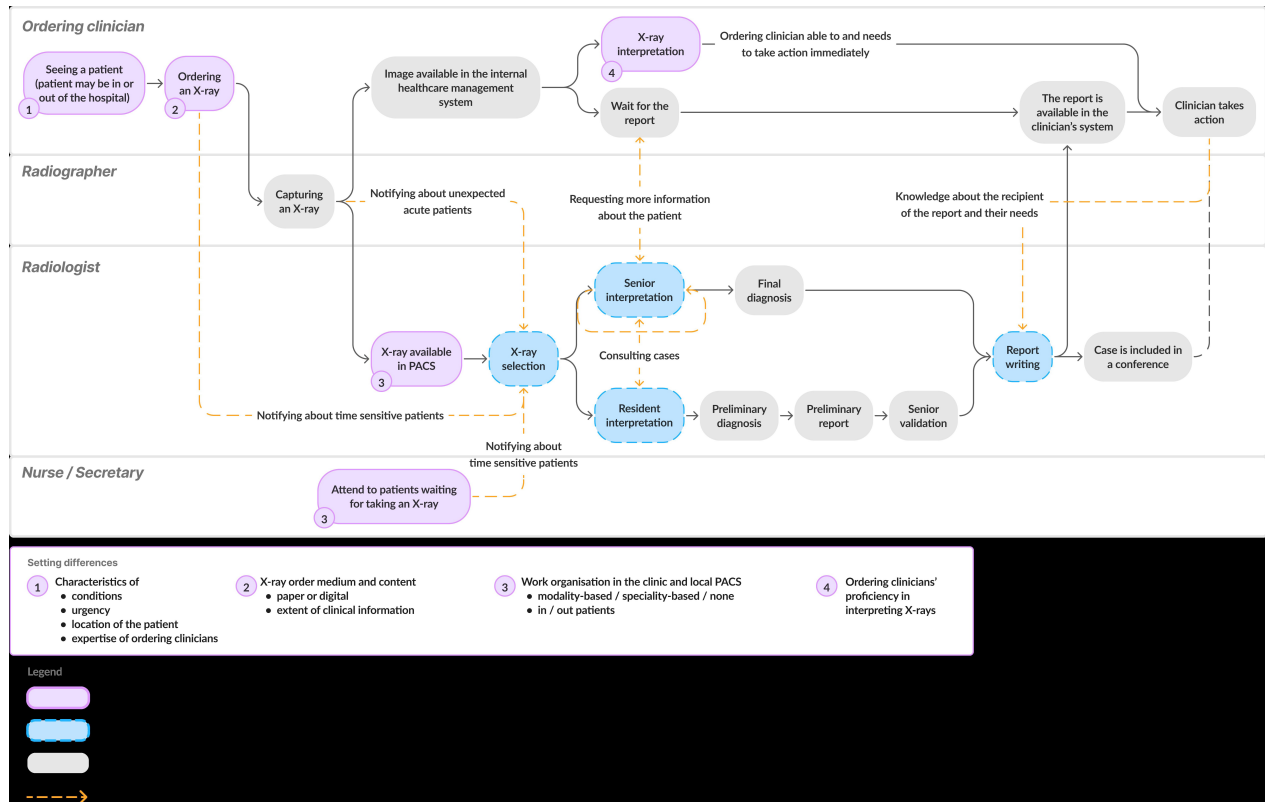


Figure 6: Assignment Figure 1 from the Assignment 2 PDF: observed radiology workflow with formal and informal coordination paths

Source: Provided in Assignment 2 brief as Figure 1 (Figur1-assignment2.jpg).

## B.5 Invisible Work Material (Notebook Part 7)

Table 12: Activities logged in the event log (12 unique activities)

| Logged activity set A   | Logged activity set B          |
|-------------------------|--------------------------------|
| Case selected           | Order placed                   |
| Critical communication  | Patient selected from worklist |
| Image acquisition start | Preliminary report created     |
| Image acquisition end   | Report distributed             |
| Images uploaded to PACS | Report finalised               |
| Order accepted          | Report validated and approved  |

Table 13: Invisible-work activities from workflow diagram not represented explicitly in log

| Activity from diagram                        | Why not logged?                                                 | Facet                | Consequence                                                          | Capture option                       |
|----------------------------------------------|-----------------------------------------------------------------|----------------------|----------------------------------------------------------------------|--------------------------------------|
| Radiologist-to-radiologist consultation      | Mostly verbal/ad hoc interaction outside structured capture     | Human, Task          | Collaborative interpretation work is under-represented               | Optional lightweight consult event   |
| Clarification call to ordering clinician     | Usually done by phone outside RIS/PACS event schema             | System, Source       | Communication-delay attribution becomes incomplete                   | Linked communication event type      |
| Informal reprioritisation and handoff        | Queue coordination is locally negotiated rather than formalised | Task, Human          | Bottleneck and fairness analyses miss practical prioritisation logic | Priority-change event with rationale |
| Cross-team coordination on patient readiness | Interdepartmental coordination spans systems and roles          | Source, System, Task | Waiting-time drivers become invisible in case timeline analysis      | Interdepartment status event         |

## B.6 Textbook Figures Referenced in the Report

The figures below are included to support direct references in Part I and Part II. To separate them from assignment-produced figures, each caption explicitly preserves the textbook figure number.

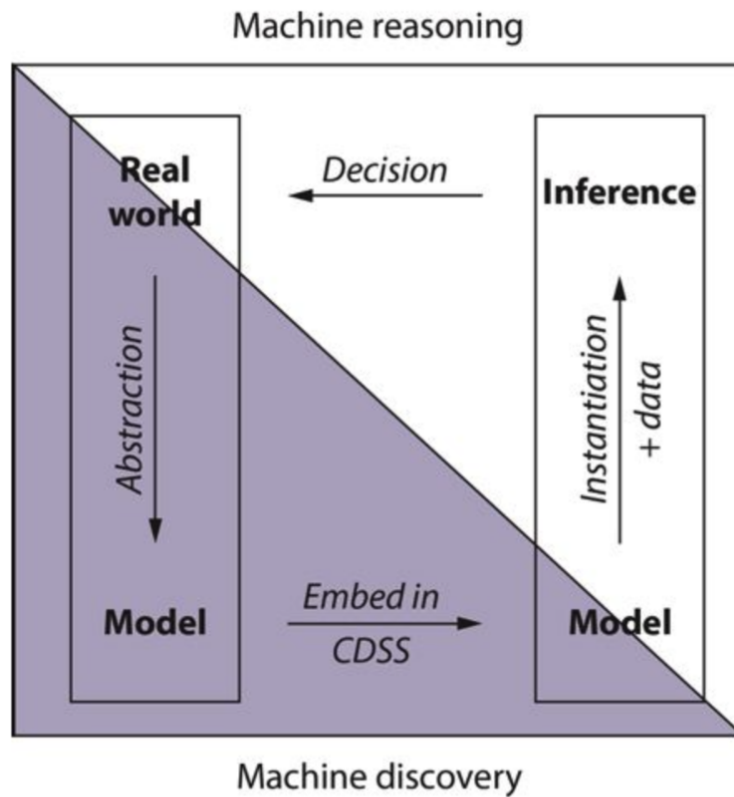
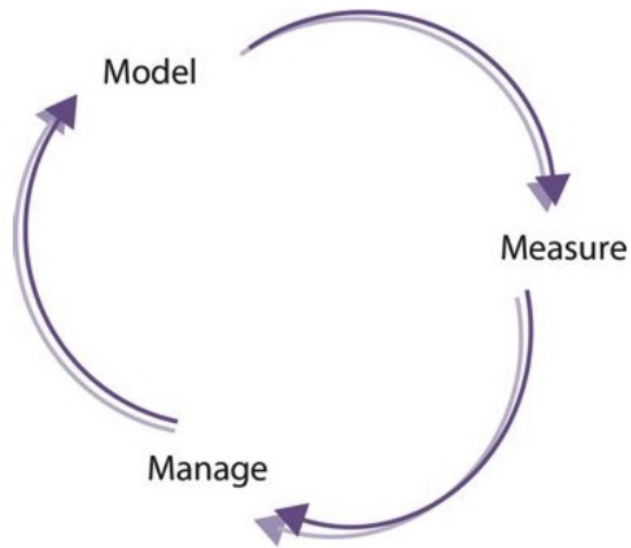


Figure 27.1 Computer reasoning depends first on acquiring models that record our understanding of how the world works and then applying those models in a decision support system to arrive at conclusions shaped by the data specific to a given decision task.

Figure 7: Textbook Figure 27.1 from *Guide to Health Informatics*: computer reasoning cycle from model construction through use

Source: *Guide to Health Informatics* (2026a). Reproduced from course compendium reading.



**Figure 9.1** The model-measure-manage cycle. When the outcome of a management decision is fed back into another cycle, a feedback control loop is formed.

Figure 8: Textbook Figure 9.1 from *Guide to Health Informatics*: model-measure-manage cycle  
Source: *Guide to Health Informatics* (2026b). Reproduced from course compendium reading.

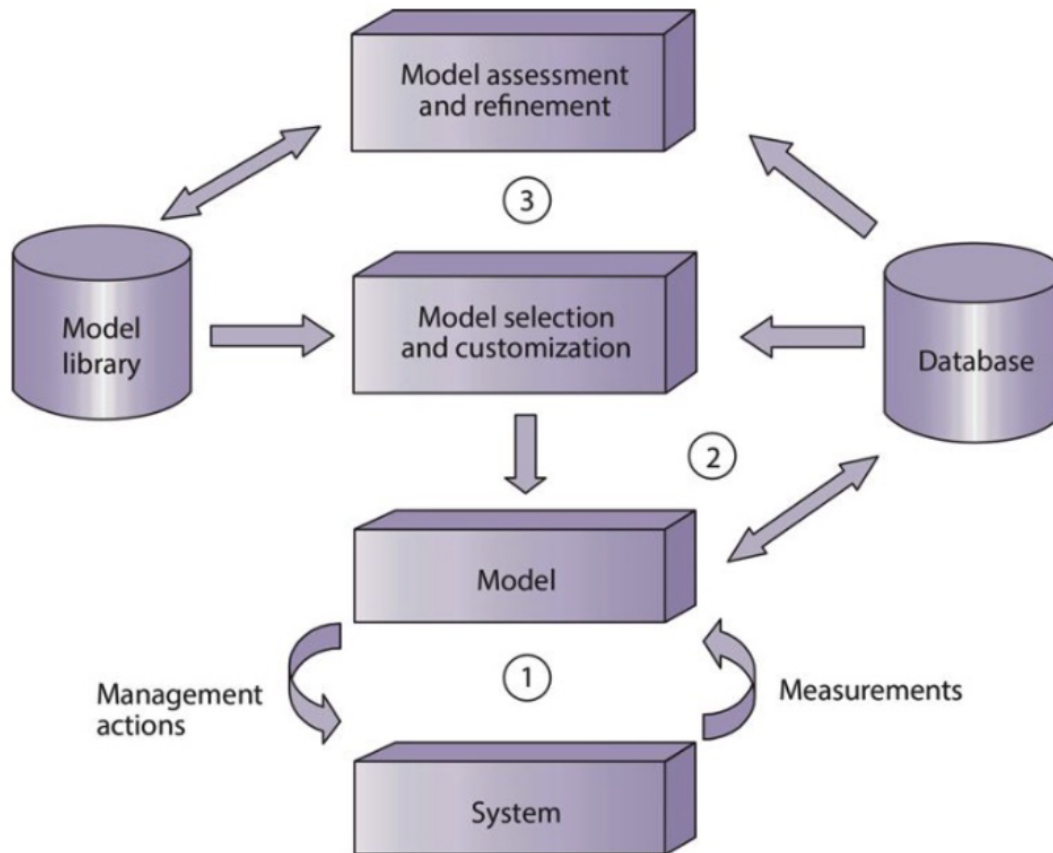


Figure 9.2 The three-loop model. Three separate information cycles interact within the health system. The first and second cycles select and apply models to the management of specific systems. The third information cycle tries to assess the effectiveness of these decisions and improve the models accordingly.

Figure 9: Textbook Figure 9.2 from *Guide to Health Informatics*: three-loop model  
 Source: *Guide to Health Informatics* (2026b). Reproduced from course compendium reading.

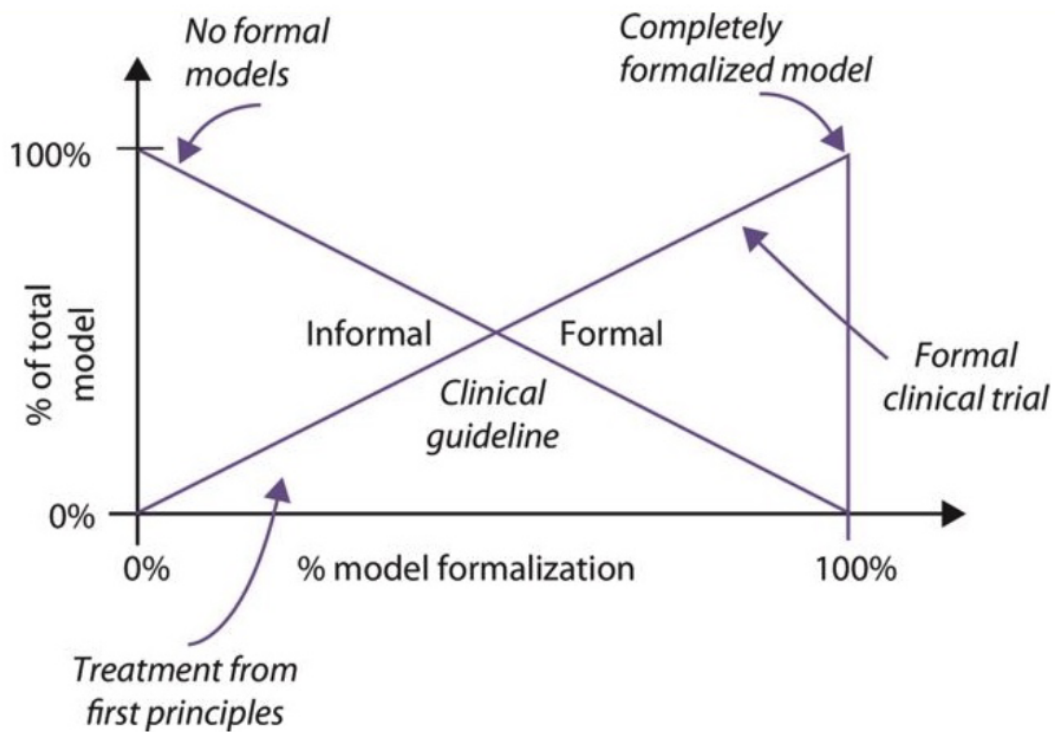


Figure 9.3 There is a continuum of possible formalization of information systems. Formal and informal models combine together to cover the total modelling needs of a given problem. The degree of formalization depends on the type of problem being addressed.

Figure 10: Textbook Figure 9.3 from *Guide to Health Informatics*: formalization continuum  
 Source: *Guide to Health Informatics* (2026b). Reproduced from course compendium reading.

## References

- Guide to Health Informatics (Course Compendium) (2026a). *Guide to Health Informatics: Selected Chapter 27*. PDF. Course reading PDF provided in NDAB20010U.
- (2026b). *Guide to Health Informatics: Selected Chapters 9 and 27*. PDF. Course reading PDF provided in NDAB20010U.
- Zajac, H. D. (Feb. 2026). *Sundhedsdata og interoperabilitet: Methods of knowing + Health-care data*. PDF. Course slides, NDAB20010U Health Data and Interoperability, University of Copenhagen.